

SegmentGDA

Training-Free Few-Shot Segmentation via Gaussian Discriminant Analysis on Vision Foundation Features

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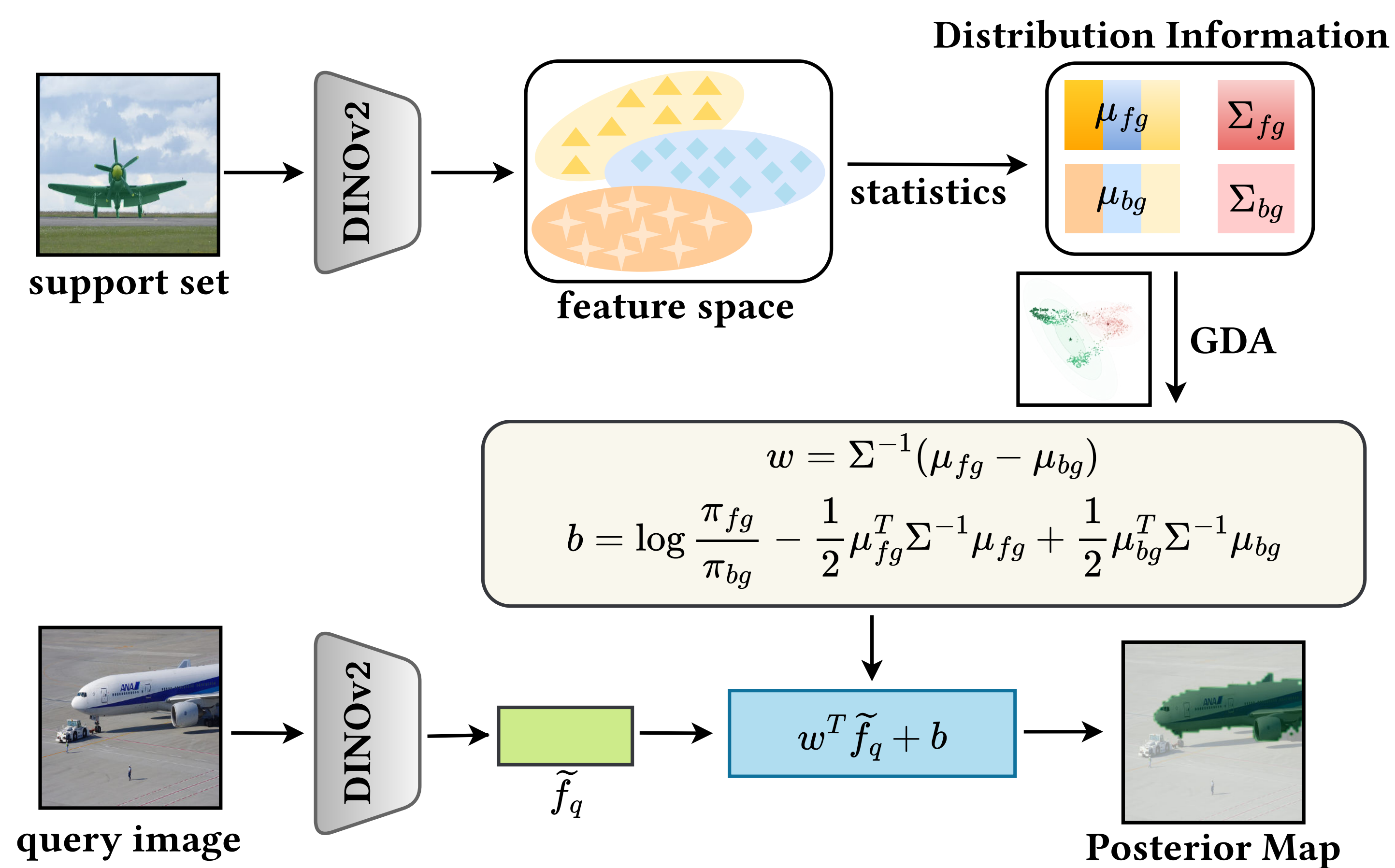
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01 Motivation

Few-shot segmentation transfers a target concept from a few annotated support images to a new query.

Local patch matching can miss object parts and confuse background clutter.

03 Method overview



04 Results across four benchmarks

Reported mIoU (%) 1-shot and 5-shot evaluation

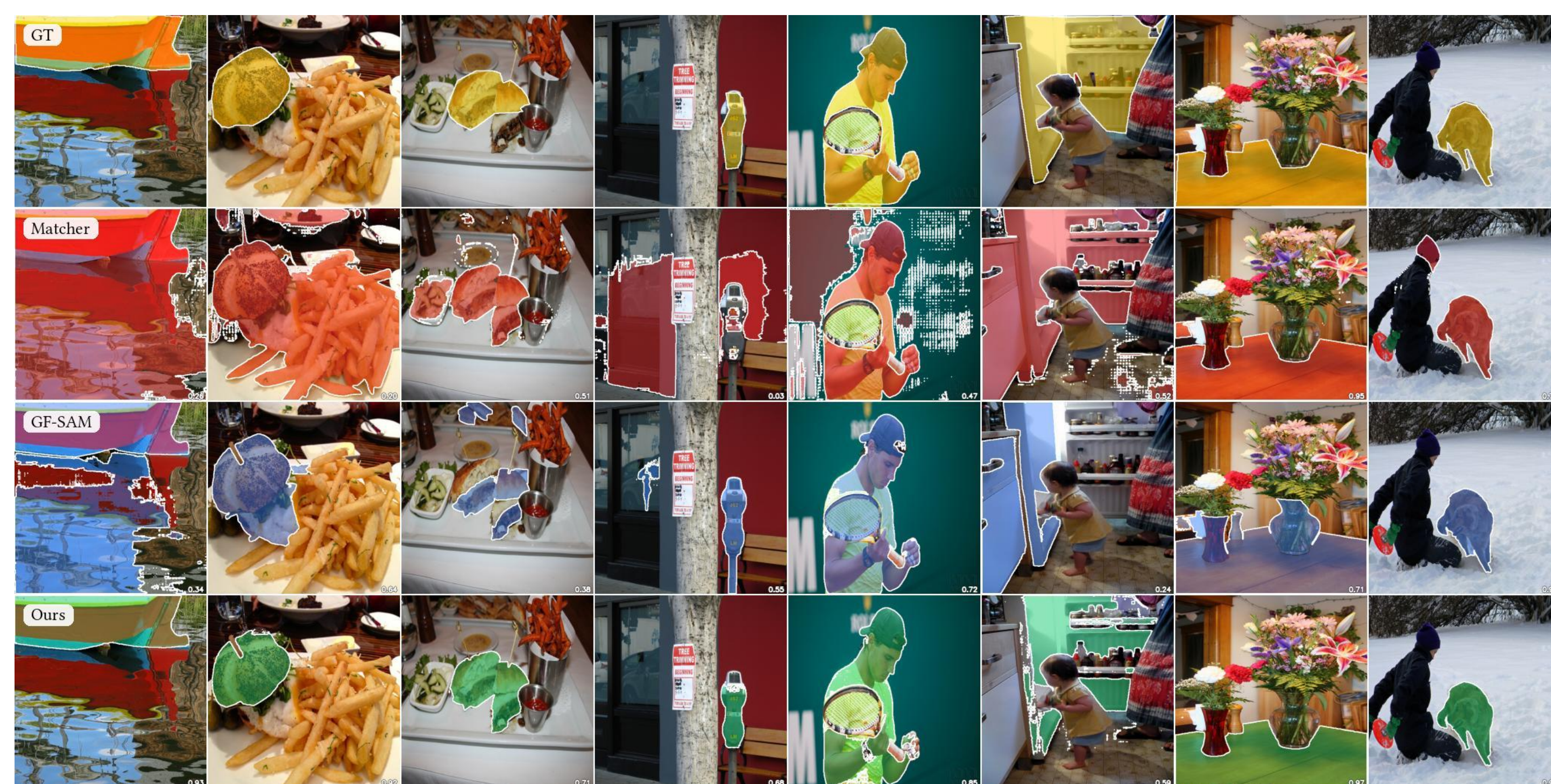
Dataset	1-shot	5-shot	Gain (pp)
COCO-20 ⁱ	42.1	62.5	+20.4
FSS-1000	82.9	89.8	+6.9
LVIS-92 ⁱ	26.2	41.8	+15.6
PASCAL-5 ⁱ	59.5	82.2	+22.7

PASCAL-5ⁱ, 5-shot: 82.2 vs. Matcher 74.0 and GF-SAM 82.6.

Largest gains arise with more support images. One-shot estimates remain less reliable.

06 Qualitative comparisons

COCO-20ⁱ

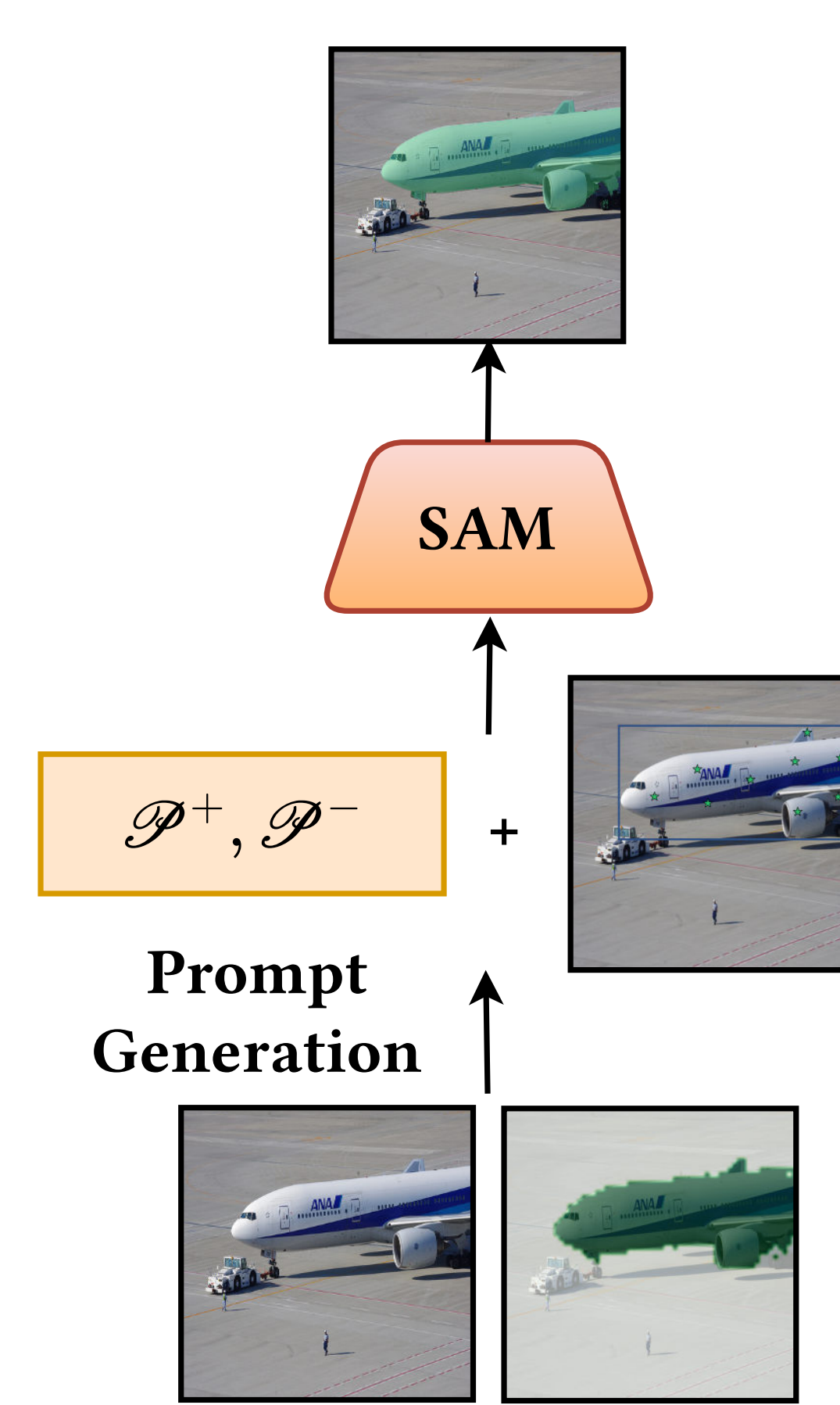


Rows: ground truth / Matcher / GF-SAM / SegmentGDA. Selected 5-shot episodes from the paper.

02 Core idea

Model foreground and background as distributions in frozen foundation features.

Use Bayesian posteriors to generate reliable SAM prompts, without task-specific training or fine-tuning.



1 Frozen features

DINOv2 ViT-L/14 extracts dense features. Apply a signed square-root transform and L2 normalization.

2 Gaussian inference

Estimate two class means and a shared, ridge-regularized covariance. Compute each query patch's foreground posterior.

3 Posterior-guided SAM

Sample spatially diverse positive and negative points. Refine errors iteratively with frozen SAM ViT-H.

05 Ablation

COCO-20ⁱ, 5-shot mIoU (%)

Point-wise matching	53.6
Mean-only Euclidean	59.9
Shared diagonal covariance	60.8
Shared full covariance	62.5

+8.9 pp

over point-wise matching in the controlled ablation.

FSS-1000



TAKEAWAY

Global feature statistics enable training-free mask transfer, with strong gains from additional support examples.

